

4D Doodling: Free Creation of Shape-Changing Decoration with A 3D Printing Pen

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ABSTRACT

4D printing thermoplastic technology has become a common method for fabricating shape-changing interfaces, however, the fully automated 3D printing method also limits user's creative participation in the fabrication process. We present a 4D Doodling method which allow users to create deformable daily items in a free-style way using a 3D printing pen. In the doodling experiment, we summarized the experience of manual printing and provided a practical method to support the user's creation, which can be a way to increase humanity for 4D printing technology.

CCS CONCEPTS

• **Human-centered computing > Human computer interaction (HCI);**

KEYWORDS

4D printing, 3D pen, shape-changing interfaces

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1 INTRODUCTION

As a new way of manufacturing, 4D printing technology can save printing time, printing materials, and packaging space. Based on the material's shape-changing property, researchers developed various methods of 4D printing in order to achieve a specific deformation. 4DTexture [1] used the deformation characteristics of PLA to design fashion accessories and functional mechanical components. 4DMesh Use 4D printing to produce customized furniture and other personalized products [2]. Both of them require 3D printers, specific printing software and background knowledge of 3D printing and design. Nowadays, many people use 3D printing pens for children's science education and artistic manual creation [3].

In this paper, we present a new 4D printing method, 4D Doodling, which combines 4D printing technology with a low-cost 3D printing pens. The contribution of 4D Doodling includes: a) explore the potential of hand-painted 4D printing and increase the possibilities of creation; b) enhance the user's sense of participation in creation and the convenience of printing compared to 4D printing with printers; c) be applied to many fields such as children's education and customization.

We test the basic printing effect of 4D Doodling and present the primitive structure design with different printing patterns. In the application, we combined the primitive structure to design different forms of flowers and pinwheels suitable for children's education, as well as some personalized accessories.

2 BASIC DEFORMATION TEST

We refer to previous studies on 4D printing deformation technique and use the thermoplastic material PLA to conduct the basic print test. We controlled other variables and test the number of layers, aspect ratios, and angles (Figure 1). The printing plate will also have an impact on the experiment result. Cutting mat is used as the printing plate, and the speed of the printing pen is adjusted to the maximum. After printing, the cooled and fixed model is put in 85 °C water to trigger the deformation.

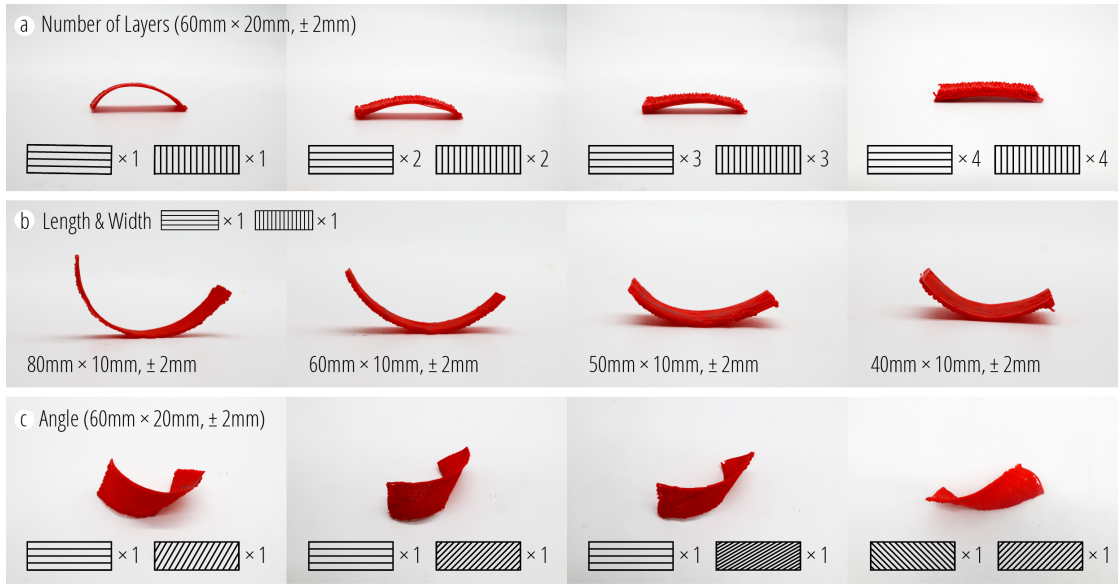


Figure 1: Other Variables Controlled, the Experiment Result of Number of layers, Aspect Ratios, and Angles

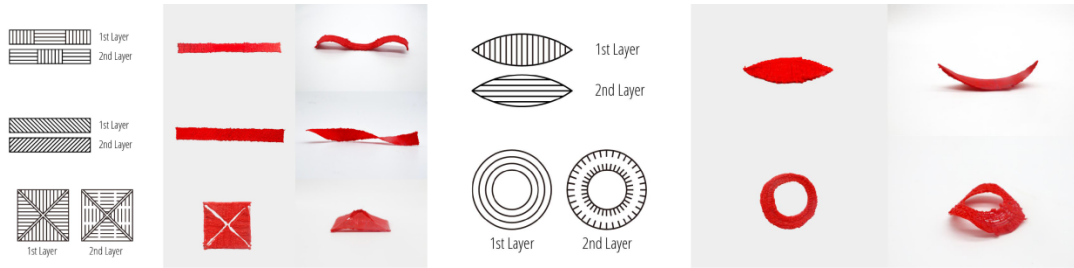


Figure 2: Primitive Deformation Library

The experiment result shows that the fewer the layers, the more the material bends after triggering; in the aspect ratio test, when ratios are close, the shrinkage will cancel each other, and the result is not as obvious as the sample with a large aspect ratio; in the angle test, the order of bending angle from small to large is 30° , 45° , 60° . Additionally, when the two layers are 45° and -45° , the sample will rotate and deform.

We conducted a series of deformation tests, and collected in the primitive structure library (Figure 2). We designed the shape and printing path of the basic unit. The deformation primitives divide into wavy, spiral, square, leaf shape, and ring.

3 APPLICATION

According to the basic deformation structure library obtained from our experiment, we designed flowers and pinwheels that children can doodle. Children's spatial cognition ability needs to be cultivated, and it can be cultivated in the process when children think about what kind of deformation different the printing paths will bring. Children can learn the basic components first, and then combine the components to form interesting products. 4D doodling can

also be combined with the physics. When designing a pinwheel, children will explore what kind of deformation will perform better and learning while playing. Users will have greater interest in the process of using 4D Doodling, participate more, and be more creative.

4D Doodling can quickly print out unique wearable products. It can also be applied to the transformation of other items in life, such as small hooks, storage boxes, etc. Traditional 3D printed models are more difficult to print on curved surfaces, while 4D Doodling allows users to design interesting and creative accessories, and saves printing time.

4 CONCLUSION AND FUTURE WORK

In this paper, we verified the feasibility of 4D printing with 3D printing pens, and showcased two fields of application using our primitive deformation method. In future work, we will explore more complex structures and designs to broaden the application scenarios.



Figure 3: Design Flowers with 4D Doodling

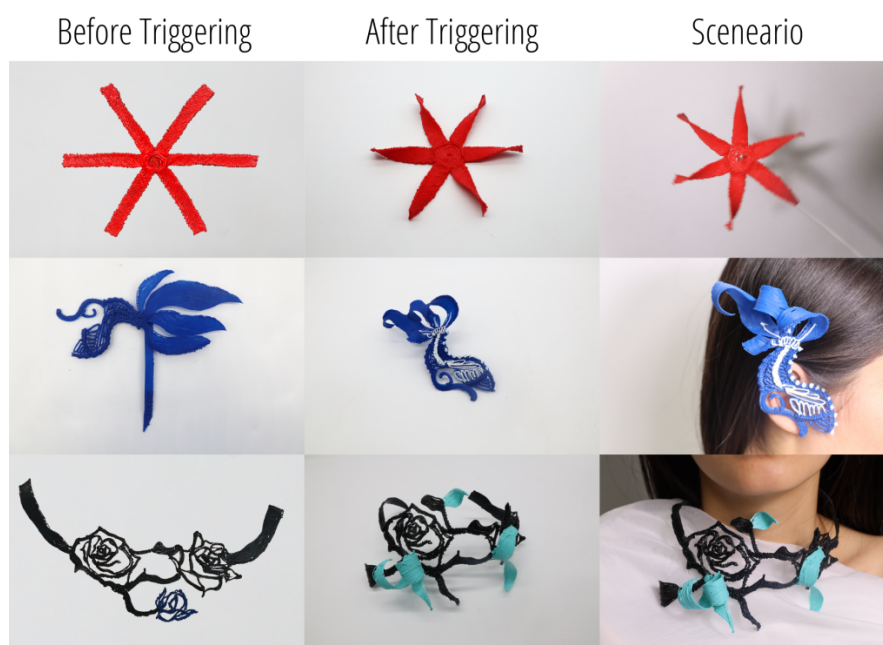


Figure 4: Making Pinwheels Customized Accessories with 4D Doodling

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REFERENCES

- [1] Lingyun Sun, Jiaji Li, Yu Chen, Yue Yang, Ye Tao, Guanyun Wang, and Lining Yao. 2020. 4DTexture: A Shape-Changing Fabrication Method for 3D Surfaces with Texture. In Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems (CHI EA '20). Association for Computing Machinery, New York, NY, USA, 1–7. DOI:<https://doi.org/10.1145/3334480.3383053>
- [2] Guanyun Wang, Humphrey Yang, Zeyu Yan, Nurcan Gecer Ulu, Ye Tao, Jianzhe Gu, Levent Burak Kara, and Lining Yao. 2018. 4DMesh: 4D Printing Morphing Non-Developable Mesh Surfaces. In Proceedings of the 31st Annual ACM Symposium on User Interface Software and Technology (UIST '18). Association for Computing Machinery, New York, NY, USA, 623–635. DOI:<https://doi.org/10.1145/3242587.3242625>
- [3] Haruki Takahashi and Jeeun Kim. 2019. 3D Pen + 3D Printer: Exploring the Role of Humans and Fabrication Machines in Creative Making. Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems. Association for Computing Machinery, New York, NY, USA, Paper 295, 1–12. DOI:<https://doi.org/10.1145/3290605.3300525>
- [4] Ya-Ting Yue, Xiaolong Zhang, Yongliang Yang, Gang Ren, Yi-King Choi, and Wenping Wang. 2017. WireDraw: 3D Wire Sculpturing Guided with Mixed Reality. Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems. Association for Computing Machinery, New York, NY, USA, 3693–3704. DOI:<https://doi.org/10.1145/3025453.3025792>
- [5] Narjes Pourjafarian, Marion Koelle, Bruno Fruchard, Sahar Mavali, Konstantin Klamka, Daniel Groeger, Paul Strohmeier, and Jürgen Steimle. 2021. BodyStylus: Freehand On-Body Design and Fabrication of Epidermal Interfaces. Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems. Association for Computing Machinery, New York, NY, USA, Article 504, 1–15. DOI:<https://doi.org/10.1145/3411764.3445475>
- [6] Amit Zoran, Roy Shilkrot, and Joseph Paradiso. 2013. Human-computer interaction for hybrid carving. In Proceedings of the 26th annual ACM symposium on User interface software and technology (UIST '13). Association for Computing Machinery, New York, NY, USA, 433–440. DOI:<https://doi.org/10.1145/2501988.2502023>

- [7] Carlos E. Tejada, Raf Ramakers, Sebastian Boring, and Daniel Ashbrook. 2020. AirTouch: 3D-printed Touch-Sensitive Objects Using Pneumatic Sensing. Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems. Association for Computing Machinery, New York, NY, USA, 1–10. DOI:<https://doi.org/10.1145/3313831.3376136>
- [8] Guanyun Wang, Ye Tao, Ozguc Bertug Capunaman, Humphrey Yang, and Lining Yao. 2019. A-line: 4D Printing Morphing Linear Composite Structures. Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems. Association for Computing Machinery, New York, NY, USA, Paper 426, 1–12. DOI:<https://doi.org/10.1145/3290605.3300656>
- [9] Lingyun Sun, Ziqian Shao, Danli Luo, Jianzhe Gu, Ye Tao, Lining Yao, and Guanyun Wang. 2020. FabricFit: Transforming Form-Fitting Fabrics. In Adjunct Publication of the 33rd Annual ACM Symposium on User Interface Software and Technology (UIST '20 Adjunct). Association for Computing Machinery, New York, NY, USA, 99–101. DOI:<https://doi.org/10.1145/3379350.3416198>